Contrastive Acoustic Embeddings for Optimal Source Selection in Low-Resource Phoneme Recognition

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Abstract

This paper addresses the challenge of low-resource phoneme recognition by introducing a contrastive learning-based method for optimal source language selection. We demonstrate that a pretrained multilingual contrastive model can identify effective source languages for a target low-resource language by computing the cosine similarity between their acoustic embeddings. Using the top candidate languages for initial pretraining, the model can then be adapted to the target language with as little as 15 minutes of transcribed speech. Experimental results show that our approach achieves an average improvement of approximately 25.9% in phoneme error rate in best case compared to directly fine-tuning Wav2Vec 2.0 models. Furthermore, we validate the effectiveness of our embedding-based selection strategy by showing a positive correlation between a source language's embedding similarity and the resulting recognition performance for the target language.

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Index Terms: speech recognition, low resource language, crosslingual processing, source language selection

1. Introduction

Thousands of languages are spoken worldwide, yet only a limited number are well-documented with sufficient data for training robust speech and language models. A central challenge in the field is to transfer knowledge from these data-rich languages to the vast number of low-resource languages. Although natural language processing (NLP) research has proposed numerous methods [1] for low-resource tasks in text level, building an applicable Automatic Speech Recognition (ASR) system remains particularly difficult due to the fundamental lack of transcribed speech data and linguistic resources. A natural and widely adopted solution is to transfer knowledge directly from well-resourced (source) languages to assist low-resource (target) languages, enabling models to leverage existing acoustic and linguistic knowledge rather than learning from scratch with minimal data. This principle of cross-lingual transfer underpins much of the progress in low-resource speech recognition. Early research primarily on developing methods to effectively acquire and combine knowledge from multiple source languages [2, 3, 4, 5, 6, 7]. These studies often selected source languages based on practical availability or broad typological diversity, with less emphasis on a systematic or optimal selection strategy for a specific target language.

A significant shift occurred with the advent of powerful self-supervised architectures like Wav2Vec 2.0 [8] and, more importantly, its multilingual counterpart XLSR[9], which is pretrained on speech from 53 languages . These large foundational models already encode a vast, shared acoustic representation.

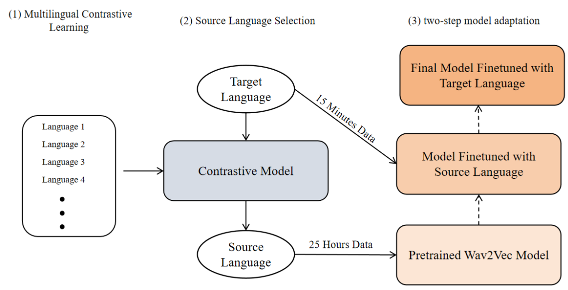


Figure 1: *Overview of the proposed three-stage framework for low-resource phoneme recognition: (1) multilingual contrastive embedding learning, (2) cosine similarity-based source language selection, and (3) two-step model adaptation.*

Consequently, the core research challenge has evolved: rather than seeking to aggregate more knowledge from numerous languages, the priority now is to efficiently guide adaptation or fine-tuning for a specific low-resource target.

Recent research addressing this challenge has primarily followed two distinct trends. The first trend focuses on data augmentation, exploring various techniques to generate or augment limited low-resource data to improve the fine-tuning of pretrained models like Wav2Vec 2.0 [10, 11, 12]. The second, more relevant trend focuses on systematic source language selection, moving beyond arbitrary or family-based choices. Here, research seeks to identify which pre-trained languages provide the most beneficial prior knowledge. For instance, some studies select languages from the same linguistic family [13], while others leverage phonological similarity [14]. Further methods employ corpus-based similarity metrics, such as comparing phoneme frequency distributions after converting text to the International Phonetic Alphabet (IPA) [15].

In this work, we address the low-resource ASR challenge by focusing on the phoneme recognition task, which presents a more tractable entry point than full word or sentence recognition. Since phonemes are frequently shared across languages, this granularity enables efficient acoustic knowledge transfer from a well-resourced source language to a low-resource target. Building on existing selection strategies, we identify the optimal selection of source languages as the critical open problem for maximizing the performance of this transfer.

We therefore propose a novel method for optimal source language selection using a pretrained multilingual contrastive model to improve low-resource phoneme recognition with limited training samples. Our approach calculates the cosine simi-

Table 1: *Top Five Source Languages by Cosine Similarity for Each Target Language*

Target	1st (Score)	2nd (Score)	3rd (Score)	4th (Score)	5th (Score)
Maltese	Turkish (62.4)	Romania(61.5)	German (61.3)	Italy (60.7)	Esperanto (60.2)
Afrikaans	Dutch (76.5)	German (72.1)	English (68.9)	Welsh (65.6)	Turkish (63.1)
Danish	Dutch (70.9)	German (70.1)	Turkish (66.3)	English (66.0)	French (62.4)
Kyrgyz	Bashkir (76.7)	Uyghur (76.4)	Uzbek(69.6)	Turkish (68.3)	Tatar (57.8)
Turkmen	Bashkir (72.9)	Uyghur (69.6)	Turkish (67.8)	Uzbek (66.4)	Arabic (54.5)
Kazakh	Bashkir (78.9)	Uyghur(72.2)	Uzbek (69.3)	Turkish (67.2)	Arabic (61.6)
Slovak	Czech (72.8)	Ukrainian (71.3)	Polish (70.4)	Belarus (70.1)	Russia (69.7)
Indonesian	Czech (65.9)	Catalan (63.8)	Arabic (62.1)	Urdu (61.8)	Basque (61.6)

larity between the acoustic embeddings of a target low-resource language and a set of high-resource candidate languages. The candidate with the highest similarity score is selected as the optimal source language for cross-lingual transfer.

To demonstrate the generality of our method, we conduct experiments on target languages spanning diverse language families, including Semitic, Germanic, Turkic, Slavic, and Malayo-Polynesian languages. Our results show that fine-tuning a model first on our selected source languages yields a significant improvement, reducing the Phoneme Error Rate (PER) by 25.9% compared to directly fine-tuning the Wav2Vec models on the limited target language data alone. Furthermore, we find a clear positive correlation between a candidate language's embedding cosine similarity and the final phoneme recognition performance for the target language. This correlation validates the effectiveness of our embedding-based selection strategy.

2. Methods

Our proposed framework for low-resource language phoneme recognition follows a three-stage pipeline, as illustrated in Figure 1. First, a multilingual contrastive learning model is trained to project speech segments into a shared embedding space, where proximity between embeddings corresponds to linguistic similarity. Second, for a target language, a small audio sample is encoded by this model. The optimal source language is selected as the candidate language whose embeddings exhibit the highest average cosine similarity to the target's embeddings. Finally, transfer learning is performed in two sequential steps: a pre-trained Wav2Vec 2.0 model is adapted using 25 hours of data from the selected source language, and subsequently fine-tuned on only 15 minutes of data from the target language. Details of each stage are as follows.

2.1. Contrastive Model

Contrastive learning has been a foundational technique in computer vision for many years[16, 17, 18]. The standard framework operates as follows: a data sample is chosen as an anchor. Positive examples (samples similar to the anchor) and negative examples (dissimilar samples) are then selected. During training, the model learns to pull the embeddings of positive examples closer to the anchor's embedding while pushing the embeddings of negative examples farther away.

We adapt this framework for phoneme representation. For training, we select languages from the Common Voice[19] 22.0 dataset that have at least 25 hours of transcribed speech. Using a supervised setup, where speech from the same language are treated as positives and speech from different languages as negatives, we train a model to learn language-discriminative acoustic embeddings.

The training objective is defined by the contrastive loss function shown in Equation 1. In this formulation, i denotes an anchor language sample, j iterates over its positive samples, and k iterates over negative samples. $\mathbb{I}$ represents the set of all training samples in a batch. $P(i)$ is the set of positive examples (speech segments) belonging to the same language as anchor i, and $A(i)$ is the set of all negative examples from other languages. The function $sim(\cdot,\cdot)$ computes the cosine similarity between two embedding vectors, and τ is a temperature scaling parameter.

$$\mathcal{L} = \sum_{i \in \mathbb{I}} \frac{-1}{|P(i)|} \sum_{j \in P(i)} \log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\mathrm{sim}(z_i, z_k)/\tau)} \quad (1)$$

2.2. Finding Source Language

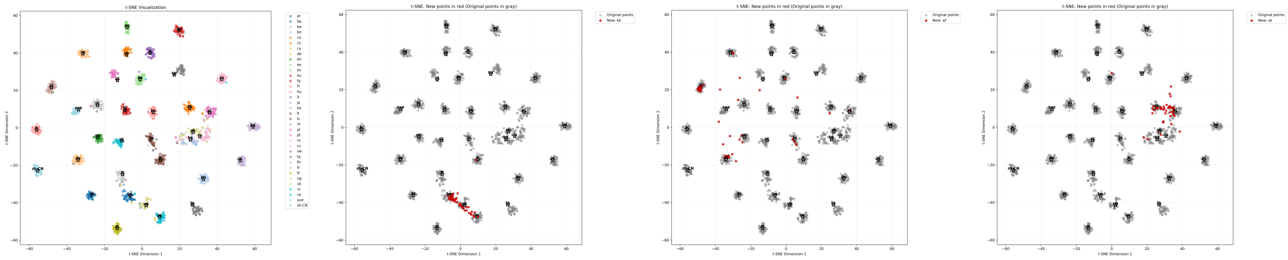
Once the contrastive model has been pretrained, we compute the similarity between a target low-resource language and each candidate source language in our pool. For every candidate language, we randomly sample N audio segments from its available training data and extract their normalized embeddings using the pretrained contrastive model. Similarly, for the target language, we draw a small set of samples (M) from its limited training data and obtain their corresponding embeddings. The similarity between the target language L_t and a candidate source language L_s is quantified as the average pairwise cosine similarity between their respective embedding sets. $\mathbf{z}_t = \{\mathbf{z}_t^{(1)}, ..., \mathbf{z}_t^{(M)}\}$ represents M normalized embeddings from the target language, and $\mathbf{z}_s = \{\mathbf{z}_s^{(1)}, ..., \mathbf{z}_t^{(N)}\}$ denotes Nnormalized embeddings from the candidate. The average cosine similarity $S(L_t, L_s)$ is computed as:

$$S(L_t, L_s) = \frac{1}{M \cdot N} \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{\mathbf{z}_t^{(i)} \cdot \mathbf{z}_s^{(j)}}{|\mathbf{z}_t^{(i)}|, |\mathbf{z}_s^{(j)}|} \quad (2)$$

After computing $S(L_t, L_s)$ for all candidate languages, we rank them in descending order of similarity. The top-ranked candidate is selected as the optimal source language for subsequent cross-lingual adaptation.

2.3. Fine-tuning

After selecting the optimal source language via our contrastive embedding similarity method, we employ a two-stage adaptation procedure. First, we pre-train a Wav2Vec 2.0 model using 25 hours of transcribed speech data from the selected source language. This stage allows the model to learn robust acoustic representations from a well-resourced language. Subsequently, we fine-tune this pre-trained model on just 15 minutes of transcribed data from the target low-resource language. This final



(a) *The t-SNE projection of the pre-trained contrastive embeddings onto a 2-dimention space.*

(b) *The t-SNE projection of Kazakh embeddings onto the pretrained space.*

(c) *The t-SNE projection of Afrikaans embeddings onto the pretrained space.*

(d) *The t-SNE projection of Slovak embeddings onto the pretrained space.*

Figure 2

stage specializes the model to the target language's specific phonetic inventory and acoustic characteristics. The phoneme error rate (PER) obtained after this fine-tuning stage serves as the primary evaluation metric for our approach.

3. Experiments

3.1. Experiment Setup

3.1.1. Dataset

For training the multilingual contrastive model, we used a balanced subset of 1,000 seconds of transcribed audio from each of 36 diverse languages: Arabic, Bashkir, Basque, Belarusian, Bengali, Catalan, Cantonese, Czech, Dutch, English, Esperanto, Persian, French, Georgian, German, Hungarian, Italian, Japanese, Latvian, Lithuanian, Polish, Portuguese, Romanian, Russian, Ukrainian, Spanish, Swahili, Tamil, Thai, Tatar, Turkish, Uyghur, Urdu, Uzbek, Welsh, and Chinese.

3.1.2. Target Languages

To evaluate the generalization of our method, we selected nine target languages representing five major linguistic families: Maltese (Semitic), Amharic (Semitic), Afrikaans (Germanic), Danish (Germanic), Kyrgyz (Turkic), Turkmen (Turkic), Kazakh (Turkic), Slovak (Slavic), and Indonesian (Malayo-Polynesian). While some languages (e.g., Indonesian) are not globally resource-scarce, their limited representation within the Common Voice dataset—typically comprising only a few hours of speech—warrants their treatment as low-resource in this study.

3.1.3. Phoneme Transcription

The dataset provides orthographic transcriptions. To obtain phoneme-level supervision, we converted all text to phoneme sequences using the eSpeak NG grapheme-to-phoneme toolkit [20].

3.1.4. Training Configuration

All experiments were conducted on an NVIDIA RTX 6000 GPU with 48GB of memory. For the main transfer learning stage, we pre-trained the model on the selected source language using a learning rate of 3e-4 . For the final fine-tuning on the target language, we used a smaller learning rate of 1e-5 to avoid catastrophic forgetting of the transferred knowledge.

3.2. Fingding Source Language

As described in the Methods section, we computed the cosine similarity between target language embeddings and candidate source language embeddings to identify the optimal transfer source. The similarity scores, scaled to a 0–100 range, are shown in Table 1, which lists the top three candidate languages for each target language.

The results demonstrate a key advantage of our data-driven approach. In cases where no suitable candidate from the same language family exists, or when multiple candidates from the same family are available, our similarity metric provides a clear, quantifiable basis for selecting the most acoustically relevant source. For example, for Kazakh (a Turkic language), the model prioritizes Bashkir over other Turkic candidates like Uzbek and Uyghur. Conversely, for Maltese (a Semitic language), the model identifies Turkish (a Turkic language) as the optimal match when in-family options are less suitable.

For visualization, we projected the learned language embeddings into a two-dimensional space using t-SNE. Figure 2(a) illustrates the original multilingual contrastive embeddings. Figure 2(b-d) shows projections for specific target languages: Kazakh (b), Afrikaans (c), and Slovak (d). These visualizations demonstrate how the contrastive model locates a new target language within the existing multilingual space and identifies a suitable source language based on proximity.

3.3. Evaluation

We evaluated our source selection method against two pre-trained baselines: wav2vec2-base and wav2vec2-XLSR-53. We also compared it to two alternative selection strategies: randomly choosing a source language, and intentionally selecting the candidate with the lowest similarity score (worst source).

The results, detailed in Table 2, lead to two main conclusions. First, using any source language for initial adaptation consistently improves performance over direct fine-tuning of the baseline on the limited target data. This is because the pre-trained Wav2Vec models, while powerful, are not optimized for any specific downstream task. Direct fine-tuning on very limited data (e.g., 15 minutes) often leads to instability and poor convergence; a preliminary adaptation step using a source language provides a more stable and effective starting point. Second, and more importantly, our similarity-based selection method outperforms both random and worst-source selection across all target languages. This demonstrates that not all source languages are equally beneficial. Compared to the baseline models, our method achieves an average reduction in

Table 2: *Phoneme Error Rate (PER, %) for each method across target languages. Lower values indicate better performance. Best results for each language and model are bolded.*

	Semitic	Germanic			Turkic		Slavic	Malayo	
	Maltese	Afrikaans	Danish	Kyrgyz	Turkmen	Kazakh	Slovak	Indonesian	Average
Wav2Vec2-base	39.8	26.6	49.9	34.0	40.2	28.4	36.8	32.7	36.3
random source	33.7	22.4	45.1	30.0	37.6	25.0	30.3	27.0	31.55
worst source	35.2	22.9	46.1	29.6	35.5	24.3	32.0	28.1	32.21
ours	**32.6**	**17.8**	**41.7**	**24.7**	**31.6**	**18.4**	**22.3**	**26.7**	**27.6**
Wav2Vec2-XLSR-53	31.5	19.1	42.4	22.7	30.6	24.7	28.1	27.4	28.5
random source	28.8	14.8	38.0	18.0	26.0	17.2	23.5	20.7	23.6
worst source	28.4	15.8	40.2	17.6	27.1	16.8	24.3	19.8	24.0
ours	**28.2**	**12.5**	**36.3**	**15.3**	**24.1**	**13.8**	**16.7**	**19.0**	**21.1**

Phoneme Error Rate of 23.9% for wav2vec2-base and 25.9% for XLSR-53.

3.4. Effectiveness of Source Selection

To validate that our contrastive model selects an optimal source language, we conducted a detailed analysis for several target languages tested in the experiment. Although a human might intuitively select Turkish for any Turkic target, our model evaluates multiple candidate languages from the training data, such as Bashkir, Uzbek, and Uyghur.

As illustrated in Figure 3, we observe a clear negative linear correlation between the embedding cosine similarity of a selected source language and the resulting phoneme error rate for a given target language. A statistical breakdown for the three Turkic target languages is provided in the fist three rows of Table 3 and we also report two languages from two other language families. The analysis shows strong negative Pearson correlations ($r \leq -0.91$) and high coefficients of determination ($R^2 \geq 0.83$), indicating that embedding similarity explains the majority of the performance variance. The associated p-values confirm these correlations are statistically significant. This result robustly validates our core hypothesis: selecting the candidate with the highest similarity score reliably yields the best-performing model, surpassing heuristic or uniform selection strategies.

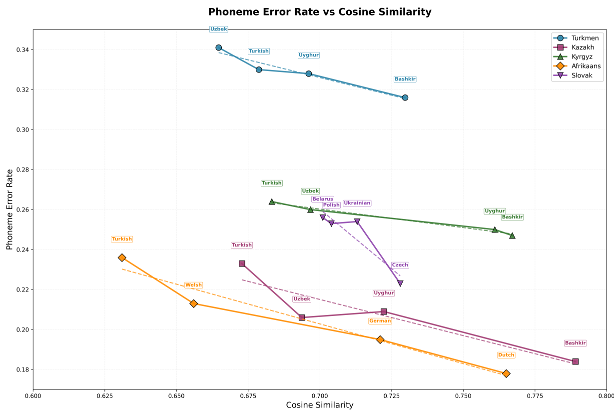


Figure 3: *PER versus cosine similarity of the selected source language. Each point represents a source–target pair, colored by target language. The dashed lines represent the linear regression fit for each target language, confirming the strong negative trend.*

Table 3: *Statistical summary of the correlation between cosine similarity and PER for each Turkic target language.*

Target Language	Pearson'r	R^2	p-value
Turkmen	-0.969	0.938	0.0314
Kazakh	-0.914	0.835	0.0865
Kyrgyz	-0.992	0.985	0.0075
Afrikaans	-0.976	0.953	0.0240
Slovak	-0.918	0.843	0.0816

4. Conclusion

This paper introduced a novel, data-driven method for optimal source language selection in low-resource phoneme recognition. By employing a pretrained multilingual contrastive model, we established a simple yet effective strategy: select the candidate language whose acoustic embeddings exhibit the highest average cosine similarity to those of the target language. Evaluated across nine languages spanning five distinct families, our approach consistently surpassed strong baselines, achieving an average reduction in Phoneme Error Rate around 25.9% in best case.

The results robustly confirm that embedding similarity serves as a reliable and quantifiable proxy for cross-lingual transferability. A key advantage of our method is its ability to identify beneficial source languages not only within the same linguistic family but also to select acoustically similar candidates across family boundaries, thereby transcending the limitations of taxonomy-driven heuristics.

Our work underscores the critical role of systematic source language selection in cross-lingual speech recognition and provides a practical, model-based framework to address it. Future work could extend this embedding-driven approach to related low-resource speech tasks, such as the recognition of tonal languages or the adaptation of end-to-end speech translation models.

5. Generative AI Use Disclosure

This research was conducted entirely by human researchers. Generative AI tools were used solely for the purpose of text polishing and grammar correction. All experiments, data analysis, interpretations, and scientific insights presented in this paper are the original work of the authors.

6. References

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